

Sasanka Ravindu Wakkumbura

BSc Information Technology Undergraduate | Software Engineer

 sasankarw@gmail.com
 +94 71 5669 231
 sasgrid.online

 in/sasankaravindu
 github.com/SasankaRW

Profile

Dynamic software engineer with experience in system integration, scalable applications, and research projects. Skilled in Python, C# and web technologies, with low-level coding expertise for hardware integration and SDK-based device connectivity.

Work Experiences

- **Software Engineer Intern, Zebra Technologies** (2025 March – 2025 September)
 - Experienced in low-level coding with SDK integration to build reliable applications for scanner connectivity.
 - Gained hands-on experience in Generative AI and Machine Learning.
 - Demonstrated strong UI/UX design skills by developing demo applications with complex user interfaces.
 - Worked with RFID and Bar code devices and fix bugs.
- **Frontend Developer (Part-time), Metacoders** (2025 Jan – 2026)
 - Built responsive front-end applications using React and modern UI libraries.
 - Created unit tests for each component to ensure reliability and maintainability.
 - Improved cross-platform performance and optimized user experience.
- **Full Stack Developer, LoonsLab** (2023 July – 2024 July)
 - Led a team in developing MedInvent Connect optimizing API performance and UI responsiveness.
 - Implemented secure authentication, API optimization, and database indexing.
 - Integrated Flutter frontend with Node.js backend.
- **Web Developer, GDG Sri Lanka** (2023 – 2024)
 - Contributed to the Google I/O website.
- **Creative Designer & Web Developer, The Ceylon Spice Hub** (2022 – 2026)
 - Designed branding materials, product labels, and digital marketing assets.
 - Designed and developed the official product website
- **Software Engineer, Freelancing** (2022 – Present)
 - Completed multiple projects across Python, web development, and research domains.
 - Delivered custom solutions ranging from automation scripts to full-stack web applications.
- **Manager, University of Moratuwa Basketball Team** (2022 – 2025)
 - Managed team operations including training schedules, logistics, and tournament organization.
 - Achieved a major milestone by winning the 2025 SLUG Championship.

Education

- **BSc in Information Technology** (2022 – 2026 July)
University of Moratuwa
Current C.GPA: 3.62
Moratuwa, Sri Lanka
- **Diploma in Cyber Security & Ethical Hacking** (2020)
CICRA Campus
Hands-on experience in penetration testing, network security, and ethical hacking methodologies.
Sri Lanka
- **G.C.E. Advanced Level** (2020)
Kingswood College
Combined Mathematics - A, ICT - A, Physics - B. Z-Score: 1.909.
Kandy, Sri Lanka

Projects

- **DeepLens (Research): Deepfake Detection via Multimodal Fusion of Frame-Level Visual Dynamics and Audio-Level Speech Cues** (Python, OpenCV, TensorFlow/PyTorch, Librosa)
Developing a multimodal deepfake detection framework combining frame-level visual analysis and audio-based speech cue detection. Built face-focused preprocessing, temporal segmentation, facial movement inconsistency analysis, and cross-modal detection pipelines for manipulated media identification.
 - **Real-Time Basketball Scoreboard Platform** (React, Firebase Realtime Database, Vercel, JavaScript)
Developed a real-time basketball scoreboard platform used in university and mercantile tournaments in Sri Lanka. Implemented multi-match support, live score synchronization, OBS-ready stream overlays, public JSON APIs, offline-first operation with automatic sync, and secure role-based access control.
 - **Ceylon Spice Hub** (React, Node.js, Express.js, MongoDB, Stripe)
Developed a full-stack e-commerce platform for a premium Sri Lankan spices company with product management, shopping cart, JWT-secured admin access, and responsive UI.
 - **OmniwareWeb** (Next.js, Express 5, MongoDB, JWT)
Engineered a full-stack e-commerce system with REST APIs, role-based authentication, rate limiting, custom PC build workflows, and MongoDB schema architecture. Deployed on Vercel.
 - **Island Safaris Sri Lanka** (Next.js App Router, TypeScript, Supabase, PayPal, Cloudinary)
Developed a booking and content management platform featuring dynamic pricing, multi-step reservation workflows, promo code support, and a secure admin dashboard.
 - **RFID Live Tracking Simulator** (WPF, C#, RFID, 3D Simulation)
Created a 3D simulation system for real-time object tracking using RFID tags and readers with location history visualization and monitoring capabilities.
 - **MedInvent Connect** - Group Project (Flutter, React, Node.js, MongoDB)
Led development of a healthcare management platform with mobile and web applications for clinic sessions, doctor appointments, and prescription tracking.
- For additional technical details and live demos, please refer to my portfolio.

Technical Skills

- **Web Development:** React, Next.js, Node.js, Tailwind CSS, RESTful APIs
- **Mobile Development:** Flutter, Kotlin, React Native
- **Database Management:** MongoDB, MySQL, Firebase, PostgreSQL
- **Programming Languages:** Python, Java, C, C#, Shell Scripting
- **Cloud & DevOps:** Docker, Firebase, GitHub Actions
- **Design Tools:** Figma, Photoshop, Adobe Illustrator, UI/UX Wireframing

Achievements

- **President's Scout of Sri Lanka** 2019
- Earned the highest rank in Sri Lanka Scouting.
- **Competitive Programming Achievements** - Finalist in the National Olympiad in Informatics 2019
- Placed 10th nationally in Google's Hash Code competition 2021
- **University Colors for Basketball** 2023, 2025
- Recognized for exceptional performance, contributing to the University of Moratuwa winning the SLUG Championship.

Referees

- **Dhanushka Adrian** - Technical Lead, Zebra Technologies
Email: adrian.dhanushka@zebra.com Phone: [+94 71 2090 625](tel:+94712090625)
- **Kamal Wijesekera** - Senior Program Manager, Pearson
Email: kamal.wijesekera@pearson.com Phone: [+94 71 6825 625](tel:+94716825625)